

112B Data Science and Python Programming

Homework Assignment #2

Due: 2024/4/19 12:00:00

We have already learned how to estimate the value of π or any definite integrals based on the Monte Carlo method, and how to check the statistical significance during the lectures. In this homework, we are going to simulate a classic problem called the “Monty Hall problem.” Your first assignment is to write a program that can simulate this game and estimate the probabilities of different game strategies. Your second assignment is to write a program that can check the significance of randomized trials. Enjoy it!

Problem #1: Monty Hall problem (50%)

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

Please write a program to simulate this game and estimate the winning probabilities of different game strategies.

Here is the sample code:

```
class solution1:
    """
    Solution for problem #1
    """
    def __init__(self, maxTrials):
        self.run_simulation(???)

    def gameMontyHall(self, ???):
        """Method for simulating 1 game"""
        ???

    def simMontyHall(self, numTrials):
        """
```

```

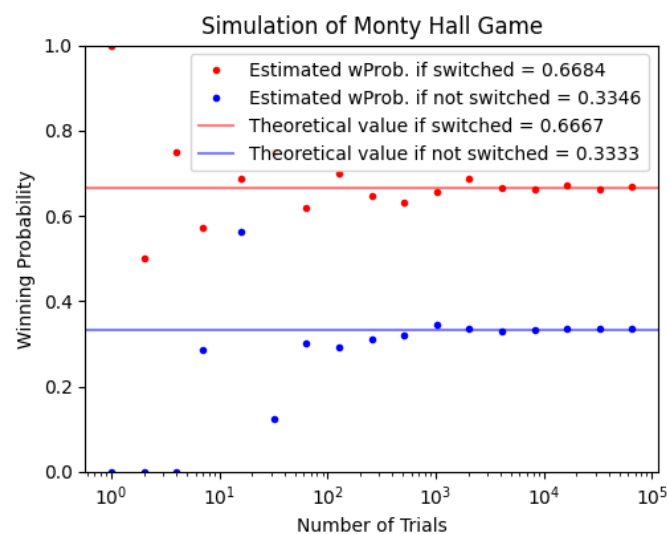
Method for simulating N game,
and return the winning probability
"""

???

def run_simulation(self, ???):
    """Method for simulation and plotting figure"""
    ???

```

If everything goes well, your code might get this result:



FRIENDLY REMINDER

You don't need to plot a figure that is exactly same as mine.

Problem #2: Checking significance (50%)

某間工廠製造的冥噤洞染劑，在製程穩定的情況下呈現一個常態分佈，每一包裝的標示重量是 94 ± 0.87 公斤。由於老闆在一場政商名流的交流賽中誇下海口要拿球棒守游擊，在比賽過程中失誤連連導致最後球隊輸球，得罪了方丈因此被多次查水表，工廠五次抽檢的結果如附檔，請問，在顯著水準 0.05 的情況下，該工廠的製程是否有出現不穩定的狀況？

Please write a program to check whether the randomized trial is statistically significant or not. The data of production inspection is in the file “**data_problem2.npz**.”

Here is the sample code:

```
class solution2:
    """
    Solution for problem #2
    """

    def __init__(self, dataDir):
        self.data = np.load(dataDir)
        ???

    def 你自己想(self,):
        return print("泰哥救救我")
```

If everything goes well, your code might get these results:

```
(nycudopcs) 0:\                               \112B\HW2>python main_hw2.py --num 2
Data 1: t = 0.925, p-value = 0.357
The production process is stable.
Data 2: t = 0.514, p-value = 0.609
The production process is stable.
Data 3: t = -1.106, p-value = 0.276
The production process is stable.
Data 4: t = 0.608, p-value = 0.546
The production process is stable.
Data 5: t = -2.617, p-value = 0.010
The production process is unstable.
```

FRIENDLY REMINDER

Any statistics-related package (e.g., `scipy.stats`)
is **STRICTLY PROHIBITED** in this homework.

Please accomplish this homework with an organized code (e.g., with main script and other script). Here is a template for your code structure:

```
112B_hw2_0123456789
├─ ooo                # Package ooo for hw2
│   ├── __init__.py
│   └─ ????.py
├─ xxx.py             # Package xxx for hw2
├─ ...                # More if necessary
└─ main_hw2.py        # Main scripts of hw2
```

The filename of your main script should be “**main_hw2.py**.”

You don’t need to follow this structure, just keep your main script clean.

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs (e.g., signature). **Your signature should be attached in your main script.** Here is the template of signature:

```
"""  
Created on Sun Aug 7 01:23:45 2022  
  
@author: Xi Winnie, student ID  
  
@collaborators: Jane Doe, her student ID  
                John Doe, his student ID  
"""
```

Please save your code as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

112B_hw2_**ID**.zip

For example,

112B_hw2_**0123456789**.zip

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “E3@NYCU” system. There will be a section for uploading your homework.